Exercise 3.1: Properties of the Propagator The propagator $K(x_1, t_1; x_0, t_0)$ is defined as the transition amplitude between two space-time points, obtained by applying the time-evolution operator $U(t_1, t_0)$ between position eigenstates:

$$K(x_1, t_1; x_0, t_0) = \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle . \tag{1}$$

We will prove a few properties of the propagator in this exercise.

a) Show that the propagator obeys Schrödinger's equation by differentiating the definition of the propagator with respect to t_1 and using $\partial U(t_1, t_0)/\partial t_1 = -\frac{i}{\hbar}\hat{H}(t_1)U(t_1, t_0)$.

b) Show that the wavefunction at time t_1 can be expressed in terms of the propagator and the initial wavefunction at time t_0 as

$$\psi(x_1, t_1) = \int dx_0 \, K(x_1, t_1; x_0, t_0)\psi(x_0, t_0), \tag{2}$$

where $\psi(x, t) = \langle x|\psi(t)\rangle$.

c) prove the following properties of the propagator:

– **Initial condition**: At equal times, the propagator reduces to the Dirac delta function

$$K(x_1, t; x_0, t) = \delta(x_1 - x_0). \tag{3}$$

– **Composition**: The propagator satisfies the composition property

$$K(x_2, t_2; x_0, t_0) = \int dx_1 \, K(x_2, t_2; x_1, t_1)K(x_1, t_1; x_0, t_0), \tag{4}$$

for any intermediate time t_1.

d) Using the free-particle propagator

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \left(\frac{m}{2\pi i\hbar t}\right)^{3/2} \exp\left(\frac{im|\mathbf{x}_1 - \mathbf{x}_0|^2}{2\hbar t}\right) \tag{5}$$

show that

$$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \langle \mathbf{p}_1 | \hat{U}(t) | \mathbf{p}_0 \rangle = \delta(\mathbf{p}_1 - \mathbf{p}_0)e^{-\frac{i}{\hbar}E_{p_0}t}. \tag{6}$$

where $E_{p_0} = \frac{|\mathbf{p}_0|^2}{2m}$ is the kinetic energy of a free particle with momentum $\mathbf{p}_0$.